ABDUL WAQAR

Artificial Intelligence Engineer

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SUMMARY

I am a BS Computer Science (Specialization in Artificial Intelligence) graduate from NED University. I have hands-on experience in Machine Learning, Deep Learning, Computer Vision, and NLP. I am skilled in Python, LLMs, and React Native, and I have completed internships at Sapphire Consulting Services, NIKUD, PIA, and PCSIR. My projects include KisaanAI, a multi-agent crop doctor for farmers, PolicyPilot, a RAG policy chatbot, a from-scratch GPT-2 implementation, a GAN-based Synthetic Data Generation system, and a LangGraph multi-agent workflow. Certified Generative AI Developer and participant at Indus AI Week 2026.

SKILLS

- **Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Image Processing, YOLO, OpenCV, PyTorch, TensorFlow/Keras, TorchVision, GANs, CNN, Transformers (BERT/GPT), Hugging Face, LLMs, RAG, LangChain, LangGraph, Agentic Workflows, Tool Calling, Data Mining
- **Languages & Frameworks:** Python, C, C++, SQL, React Native, Node.js, FastAPI, Streamlit, HTML, CSS
- **Tools & Cloud:** Git, GitHub, Firebase, Google Colab, Scikit-learn, ChromaDB, Ollama, CUDA/GPUs, MLflow/W&B, Databases (Pandas, NumPy, Database Management)
- **Core Competencies:** Problem Solving, Team Collaboration

CERTIFICATIONS AND ACHIEVEMENTS

- Generative AI Application Developer Training - NCEAC-HEC (iCodeGuru, PakAngels, Aspire Pakistan, UETIANS Lahore) | 2026
- Python Programming - edX | 2025
- Indus AI Week - NYLP | Feb 2026 - Participated in a 3-day national conference covering AI advancements, Women in AI, AI for All, and Youth in AI

EXPERIENCE

Sapphire Consulting Services - AI Engineering Intern **Jul 2026 - Aug 2026**
Karachi, Pakistan

- Built a DCGAN to generate synthetic images, addressing class imbalance for the underrepresented class in a classification dataset.
- Trained a CNN classifier on the GAN-augmented dataset, improving accuracy on the minority class.
- Developed a Streamlit app to generate new images and display the Discriminator's real/fake confidence score in real time.

NIKUD - React Native and Full Stack Intern **Jul 2025 - Aug 2025**
Karachi, Pakistan

- Developed a cross-platform mobile application using React Native and Firebase for Android.
- Built an Admin Dashboard using Node.js for real-time data visualization and user management.
- Integrated RESTful APIs to connect the mobile app with the backend dashboard.
- Implemented Firebase Authentication and Realtime Database for enhanced security and data retrieval.

Pakistan International Airlines (PIA) - React Native Intern **Aug 2024 - Sep 2024**
Karachi, Pakistan

- Developed cross-platform mobile applications using React Native and Firebase.
- Implemented real-time database integration for improved app responsiveness.
- Collaborated with senior developers to debug and optimize code, reducing app crashes.

Pakistan Council of Scientific and Industrial Research (PCSIR) - Data Analyst Intern **Sep 2023 - Oct 2023**
Karachi, Pakistan

- Performed data cleaning and preprocessing on large datasets, handling missing values and inconsistencies.
- Improved data accuracy through systematic validation and outlier detection.

- Assisted in data mining tasks to increase reporting efficiency.

PROJECTS

KisaanAI - Multi-Agent Crop Doctor | Python, FastAPI, Groq, Whisper, Urdu TTS

- Built a 5-agent system for Pakistani farmers: a leaf photo and Urdu voice note go through listener, diagnosis, research, verifier, and advisor agents and return spoken Urdu advice. Built solo for a hackathon and deployed on Render.
- Designed a verifier that checks doses against the cited source in code, blocks banned pesticides, and adds weather-aware spray warnings, so the system never invents a dose.

PolicyPilot - Employee Policy Q&A (RAG) | Python, RAG, ChromaDB, Mistral, Streamlit

- Built a RAG chatbot that answers only from company policy documents, cites the source policy, and refuses when the documents don't cover the question.
- Wrote a 27-case evaluation harness that caught retrieval and scope bugs, and added caching, retries, and query logging.

GPT-2 Implementation from Scratch | Python, NumPy

- Implemented a GPT-2 style transformer from scratch in NumPy, including tokenization, embeddings, masked multi-head self-attention, and backpropagation.
- Trained a character-level language model using AdamW, and implemented autoregressive text generation with temperature control.

Synthetic Data Generation for Face Classification | Python, PyTorch, DCGAN, CNN, Scikit-learn, Streamlit

- Built a DCGAN to generate realistic synthetic images, addressing class imbalance in a classification dataset.
- Trained a CNN classifier on the augmented dataset to improve performance on the underrepresented class.
- Built a Streamlit app to generate new images and display the Discriminator's real/fake confidence score.

Multi-Agent Blog Writer | Python, LangGraph, Groq API, FastAPI

- Built a LangGraph multi-agent workflow (planner, worker, reviewer) where the reviewer approves a draft or sends it back for fixes, capped at 2 retries.
- Added a real-time FastAPI dashboard streaming each agent step, reviewer feedback, and final status, and deployed it on Render.

Tool-Using Research Agent | Python, Groq API (GPT-OSS 120B), Tool Calling

- Built an agent that picks between web search, page fetch, and summarise tools, with a step budget to stop endless loops.
- Added source tracking and failure handling, and required every answer to use only facts from fetched pages.

Hate Speech and Cyberbullying Detection | Python, NLP, Scikit-learn, Streamlit

- Built an AI classifier for Hate Speech, Cyberbullying, or Neutral text using TF-IDF and Logistic Regression.
- Combined machine learning with rule-based keyword filtering for low-confidence cases.
- Developed an interactive Streamlit app with real-time analysis, confidence scores, and confusion matrix visualization.

EDUCATION

NED University of Engineering and Technology — **2022 - 2026**
Bachelor of Computer Science - Specialization in Artificial Intelligence — Karachi, Pakistan
CGPA: 3.17

Govt. Degree Science/Commerce College, Gulshan-e-Iqbal — **2020 - 2022**
Intermediate - 81.09% — Karachi, Pakistan

Everest Grammar High School — **2018 - 2020**
Matriculation - 82.5% — Karachi, Pakistan

LANGUAGES

- English – Intermediate
- Urdu – Native
- Sindhi – Native